SESSION 5 – LOGISTIC REGRESSION (BINARY CLASSIFICATION)

🎯 **Topic**

- Logistic Regression
- Binary Classification x
- Sigmoid Function (intuition)
- Probability & Threshold
- First Classification Model (Practical)

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1. 🎯 Learning Objectives

By the end of this session, students will be able to:

- Understand what **Logistic Regression** is
- Know why Linear Regression is **not suitable for classification**
- Understand **probability-based prediction**
- Learn **Sigmoid function (concept only)**
- Build a **binary classification model using Logistic Regression**
- Interpret predictions and probabilities

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2. 🔁 Quick Recap (Connect Previous Session)

Start class like this:

"In the previous session, we learned what classification is.
Today we will learn our first classification algorithm — Logistic Regression, which is widely used in real-world applications."

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3. 📖 What is Logistic Regression?

3.1 Definition

Logistic Regression is a supervised machine learning algorithm used for **binary classification problems**.

3.2 Simple Meaning (Most Important Line)

"Logistic Regression predicts the probability of a class and then converts it into a category."

3.3 Example

Input	Output
Study Hours	Pass / Fail
Email	Spam / Not Spam
Patient Data	Disease / No Disease

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4. ❗ Why Not Use Linear Regression for Classification?

Problem 1: Output Range Issue

Linear Regression can predict:

$$(-\infty, +\infty)$$

👉 But classification needs:

$$0 \text{ or } 1$$

Problem 2: Not Probability-Based

Linear Regression:

- Does NOT give probability

But classification needs:

- Probability (e.g., 80% chance of pass)

Teaching Line

"Linear Regression predicts values, but classification needs probabilities."

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5. 📊 Solution: Logistic Regression

Key Idea

Instead of predicting direct output:

👉 Logistic Regression predicts:

Probability(0 to 1)

6. 📈 Sigmoid Function (Core Concept)

Definition

Sigmoid function converts any value into a range between 0 and 1.

Formula (Only Show, Don't Derive)

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Output Range

Input	Output
-∞	0
0	0.5
+∞	1

Simple Explanation

"Sigmoid takes any number and compresses it into a probability between 0 and 1."

Teaching Line

"Sigmoid is the reason Logistic Regression works for classification."

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7. 📊 From Probability to Class (Threshold)

Rule

Probability	Class
≥ 0.5	1 (Yes / Pass)
< 0.5	0 (No / Fail)

Example

Probability	Output
0.8	Pass
0.3	Fail

Teaching Line

"Logistic Regression first predicts probability, then uses a threshold to decide the class."

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8. 📊 Logistic Regression Equation (Concept Only)

Then apply sigmoid: z=b0+b1x

$$y = \sigma(z)$$

Meaning

1. Linear combination
2. Apply sigmoid
3. Get probability

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9. 📌 Decision Boundary

Definition

A decision boundary is a line where probability = 0.5

Example

- Left side → Fail
- Right side → Pass

Teaching Line

"Decision boundary separates two classes."

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10. 📊 Real-Life Example (Important)

Example: Student Pass/Fail

Study Hours	Result
2	Fail
6	Pass

👉 Model learns:

"More study hours → higher probability of passing"

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11. 📊 Key Properties

✅ **Outputs probability**

✅ **Works for binary classification**

✅ **Fast and efficient**

✅ **Easy to interpret**

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12. 🎯 Interview Concepts

Q1: What is Logistic Regression?

👉 A classification algorithm that predicts probability.

Q2: Why not Linear Regression?

👉 Because it cannot limit output between 0 and 1.

Q3: What is Sigmoid Function?

👉 A function that converts values into probability.

Q4: What is threshold?

👉 A value (usually 0.5) used to convert probability into class.